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AI-driven detection and classification of diabetic retinopathy stages using EfficientNetB0

Deepak Dharrao^{1*}, Madhuri Dharrao², Shreyas Patil¹, Sangeeth Salvin¹, Prashant Ahire⁴ and Yashwant Dongre³

*Correspondence:

Deepak Dharrao

deepak.dharrao@sitpune.edu.in

¹Department of Computer Science and Engineering, Symbiosis Institute of Technology, Symbiosis International (Deemed University), Pune, Maharashtra, India

²School of Computer Applications, Pimpri Chinchwad University, Maval, Pune, Maharashtra, India

³Department of Computer Engineering, Vishwakarma Institute of Technology, Pune, Maharashtra, India

⁴Nutan Maharashtra Institute of Engineering and Technology, Talegaon Dabhade, Pune, Maharashtra, India

Abstract

Diabetic retinopathy (DR) is a major cause of preventable blindness worldwide with more than 146 million people with diabetes worldwide. Automated screening systems are crucial for early detection and thus reduction of vision loss, especially in resource-limited settings where access to specialists is limited. This paper introduces a single-architecture with lightweight deep learning framework, EfficientNetB0 with Multi-Scale Spatial Attention Gates (MSAG), to detect and classify the severity of automated DR detection and severity. The preprocessing pipeline consists of green-channel extraction, contrast-limited adaptive histogram equalization (CLAHE) and massive data augmentation to deal with imaging variability and class imbalance. The model is trained using APTOS 2019 (3,662 training, 1,928 test images) and has a quadratic weighted kappa (QWK) of 0.923 $\pm$ 0.008 and macro-averaged AUC of 0.976, which is better than the ResNet50 (QWK: 0.901) and DenseNet121 (QWK: 0.908) baselines. The grad-cam visualization analysis results in interpretable attention maps containing pathologically important areas. The lightweight architecture can achieve sub-second inference using 5.3 M parameters, which is ideal to be deployed in mobile screening as well as resource-constrained clinical settings.

Keywords Diabetic retinopathy detection, Convolutional neural network, EfficientNetB0, Attention mechanism, Quadratic weighted kappa

1 Introduction

Diabetic Retinopathy (DR) is one of the most common and severe microvascular complications resulting from diabetes mellitus. It impairs the blood vessels in the retina and when untreated may cause permanent blindness. International Diabetes Federation states that it is a disease that is not inherited. DR now covers more than 146 million individuals across the world, and there is a possibility that the number might hit 200 million in the future by 2030 [1]. DR affects more than 9.6 million individuals in the United States alone, and its economic impact is more than 15.4\$ billion in direct medical expenditures and loss of productivity annually [2]. The incidence rate of DR in diabetic patients across the world is estimated to be between 18% and more than 35% depending



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on a number of factors including regional healthcare facilities, glycemic regulation, and retinal screening frequency [3]. During the initial stages of DR patients are frequently asymptomatic and thus hard to diagnose and treat. The existing standard diagnostic process includes manual grading of color fundus photographs by ophthalmologists, which is both time-consuming and expensive and prone to inter-observer variability [4]. Moreover, the lack of trained ophthalmologists, especially in rural and underserved areas, is a significant obstacle to large-scale DR screening initiatives. As Artificial Intelligence (AI) and Deep Learning (DL) have become important in medical imaging, automated There is an increasing interest in DR detection systems [5]. Deep Convolutional Neural Networks (CNNs) have proven to be better image classification tasks, the performance of which is frequently comparable or even superior to that of human experts [6]. The models when trained appropriately can detect the less obvious retinal lesions like microaneurysms, hemorrhages, and exudates which are indicative of DR stages. Nevertheless, there are several issues in application of these systems in practical clinical settings. These contain differences in image quality, artifact existence, class imbalance, and a deficiency of model interpretability and transferability between datasets.

This study addresses critical gaps in automated screening of DR by creating a lightweight interpretable deep learning model that can be deployed to a wide range of clinical settings. We are aiming at achieving: (1) Computational Efficiency: Build a single-model framework that delivers specialist-level performance with low computation requirements that can be deployed on mobile and at the edge. (2) Clinical Interpretability: Combine attention tools and visualization features that give clinically significant lesion localization to be verified by experts. (3) Cross-Dataset Generalizability: Test model performance on multiple datasets that represent different imaging conditions and populations.

We have made the following contributions: (1) Novel Multi-Scale Spatial Attention Gate (MSAG) mechanism to improve the lesion localization of EfficientNetB0, (2) Comprehensive ablation studies to quantify the contribution of individual components, (3) Extensive external validation to show a cross-dataset generalization and (4) Clinical interpretability analysis with expert validation of attention visualizations. This research aims to develop a lightweight and efficient CNN model for automated DR detection and classification using the EfficientNetB0 architecture. EfficientNetB0 is a compound-scaled convolutional model known for achieving high accuracy while maintaining computational efficiency. It is particularly well-suited for deployment in real-time diagnostic systems and mobile healthcare platforms. In this work, the use of transfer learning, image preprocessing, and data augmentation techniques are explored to enhance the model's performance. The objective is not only to improve classification accuracy but also to reduce training complexity and inference time. The results demonstrate that the proposed approach offers a practical and effective solution for DR stage classification in clinical environments.

2 Literature review

In recent years, deep learning, ensemble methods, and hybrid architectures have seen substantial progress in the automated detection of diabetic retinopathy. The availability of high-quality datasets such as the APTOS 2019 Blindness Detection dataset has given yet another thrust to research efforts. Diabetic Retinopathy (DR) is broadly classified

into two forms: non-proliferative DR (NPDR) and proliferative DR (PDR). These are distinguished by the presence of microaneurysms, hemorrhages, and neovascularization, which are evident in retinal fundus images.

2.1 Traditional and CNN-based approaches

Early automated DR detection systems were based on handcrafted feature extraction with traditional machine learning classifiers [7]. These algorithms usually consisted of vessel segmentation, microaneurysm locations based on morphological operations, and texture analysis based on Local Binary Patterns (LBP) or Gray-Level Co-occurrence Matrix (GLCM) features, and Support Vector Machine (SVM) or Random Forest classification [8]. These methods were moderately successful (AUC: 0.75–0.85) but demanded a large amount of domain knowledge and did not generalize as well on different datasets. Convolutional neural networks (CNNs) transformed automated DR detection [9], allowing end-to-end learning on raw fundus images. The use of deep learning to detect DR on a large scale was first proposed by Gulshan et al. [10], who returned 87.0% sensitivity and 98.5% specificity with the Inception-v3 architecture. Later work showed that learning based on ImageNet-pretrained models can be used to achieve high-performance on medical imaging tasks with a small quantity of labels [11]. The recent research papers have examined different CNN models in classifying DR. According to Mohanty et al. [12], 97.3% of the results with DenseNet121 prove the potential of transfer learning. Alwakid et al. [13, 14] also enhanced the classification performance using CLAHE contrast enhancement and ESRGAN super-resolution prior to DenseNet121 to a maximum of 98.36% accuracy. Mondal et al. [15] proposed EDLDR, a combination of ResNeXt and DenseNet models trained on APTOS 2019, with an accuracy of 86.08.

Preprocessing has a dramatic effect on DR classification and in particular in retinal imaging where illumination changes, field of view and imaging device variations are typical. Green channel extraction has become a standard preprocessing method, which offers the best contrast in vascular structures and hemorrhagic lesions. Alwakid et al. [13, 14] showed that CLAHE enhancement and super-resolution methods combined to achieve DenseNet121 98.36% accuracy on APTOS 2019. Geometric augmentation (rotation, flipping, translation) simulating positioning variation between patients, and photometric augmentation (brightness, contrast, gamma correction) to deal with illumination variation are particularly designed data augmentation strategies specific to fundus imaging [16]. Further advantages of advanced methods like deformations of elasticity to model variability in the anatomy and cutout augmentation have been demonstrated to have advantages in addressing class imbalance [17].

2.2 Attention mechanisms in medical imaging

Attention mechanisms have found popularity in medical image analysis due to their capability to emphasize clinically significant parts and give interpretable visualizations [18]. In the case of DR detection, attention mechanisms are useful in localizing pathologic alterations, i.e. microaneurysms, hemorrhages, and exudates. Recent attention on transformers demonstrated promise in retinal imaging, with Karkera et al. [19] and Wei, X., et al. [20] examining transformer networks on APTOS 2019 and finding that transformers perform best at modelling local and global lesion patterns. Xu et al. [21] developed a hybrid EfficientNet-Swin Transformer that has 97% accuracy and provides visual

evidence in the form of class activation maps. These neural network hybrid structures (that consist of CNNs and attention and transformer modules) have also enhanced the localization and interpretability of lesions.

2.3 EfficientNet architectures and model efficiency

Neural architecture search with compound scaling optimization has discovered EfficientNet architectures with remarkable accuracy-efficiency trade-offs that can be utilized in medical applications with resource constraints [22]. Anand et al. [23], optimized EfficientNetB0 on 3,200 images and achieved an accuracy of almost 91%, surpassing more complex networks such as ResNet and DenseNet at a very low computational expense. Arora et al. [24] presented an ensemble model with EfficientNetB0 to classify the severity of the DR, where the ensemble had an average classification rate of about 86.5, which proves the viability of lightweight models in the clinical environment.

2.4 Ensemble methods and hybrid approaches

Ensemble learning has been useful to deal with the issue of class imbalance and enhance generalization on DR detection. Fusion strategies enhance predictive accuracy by integrating various CNN backbones such as Xception and EfficientNet-B4, which are trained on variously processed images [25]. Ishtiaq et al. [17] combined feature maps based on a CNN and local binary pattern (LBP) texture descriptors and achieved an accuracy of 98.85% on the EyePACS dataset. Also, Dharrao et al. [26, 27] explored the machine learning algorithms to classify diabetic conditions and deep learning to classify them, focusing on the significance of large datasets and methods of extracting features applied to DR detection frameworks.

2.5 Recent advances and vision-language models

Recently, lightweight multi-deep learning frameworks have been developed that perform better [28]. There is still potential in advanced preprocessing methods and multi-modal methods to develop robust DR systems capable of explaining their outputs and are clinically viable when trained on high-quality datasets such as APTOS 2019 [29].

2.6 Research gaps and positioning

Despite several advancements, there remains a number of constraints to current DR detection systems: (1) The majority of high-performing models are based on complicated ensemble structures, which cannot be deployed in the mobile environment, (2) There is a lack of external validation on a wide range of datasets and populations, (3) Inadequate consideration of interpretability requirements to clinical adoption, and (4) Incomplete ablation studies of the contributions of individual components. We fill these gaps by coming up with one lightweight architecture that contains integrated attention mechanisms, a full-fledged multi-dataset validation and a clinically validated interpretability analysis.

3 Materials and methodology

3.1 Dataset description

The Asia Pacific Tele-Ophthalmology Society (APTOS) 2019 Blindness Detection data set consists of 3,662 training images and 1,928 test images [29]. All the pictures are



Fig. 1 Sample fundus images of 5 stages of DR severity (0–4) of the APTOS 2019 data

Table 1 Diabetic retinopathy stage definitions

Stage	Characteristics
No DR	No visible abnormalities
Mild	Presence of microaneurysms
Moderate	Increased microaneurysms and hemorrhages
Severe	Extensive hemorrhages, venous beading, IRMA
PDR	Neovascularization, vitreous hemorrhage

graded by qualified ophthalmologists on a five-grade severity scale, which includes 0 (No DR), 1 (Mild), 2 (Moderate), 3 (Severe), and 4 (Proliferative DR). The images were taken in a variety of conditions with differences in the illumination, field of view and camera models as shown in Fig. 1, which simulates clinical diversity in the real world.

One of the issues is the imbalance in classes: normal cases constitute more than 40% of the data, whereas there are less than 10% proliferative cases, which complicates the task of multi-class classification.

Table 1 provides a summary of clinical criteria used to define the five stages of diabetic retinopathy. To obtain external validation, three more datasets can be used: (1) EyePACS with 35,126 images of diabetic patients in California screening programs, which are high-quality standardized imaging conditions, (2) Messidor-2 with 1,748 high-resolution images in France that were captured under standardized protocols and devices, and (3) IDRiD (Indian Diabetic Retinopathy Image Dataset) with 516 images especially curated to be used in research with extensive lesion annotation [30]. Such datasets offer varying imaging conditions, patient groups, and annotation plans, which allow the thorough evaluation of model generalizability in varying clinical settings.

3.2 Preprocessing pipeline

The preprocessing pipeline will be configured to normalize the variability of imaging and maintain pathological features that are important in the classification of DR. The green color channel of RGB fundus image offers best contrast to blood vessels and lesions. Adaptive histogram equalization (AHE) or contrast-limited AHE (CLAHE) is used to enhance local contrast, hence lesions like microaneurysms and hemorrhages become more prominent. CLAHE prevents noise amplification in homogeneous regions. To scale to the [0, 1] range or standardize to zero mean and unit variance is used as a method of normalizing pixel values. This minimizes differences in brightness and color among samples. Each image is downsized to a fixed size (224×224) to fit the size constraints of CNNs. The median filtering is used on the preprocessed images to reduce high frequency noise like salt-and-pepper artifacts.

3.3 Data augmentation strategy

Different augmentation methods are used during training to reduce the imbalance in classes and enhance model generalization. Variability in patient positioning and acquisition conditions are simulated by random rotations ($\pm 15^\circ$), flips, and translations. Brightness, contrast and gamma variations are used to simulate real-world inconsistencies in image capture. Because of the imbalance of the datasets, a class-balanced sampling is performed with a blend parameter $\alpha = 0.5$, between empirical and uniform class distributions.

3.4 Model architecture

The backbone architecture is chosen as EfficientNetB0 due to the thorough examination of the accuracy-efficiency trade-offs that are vital when deploying the architecture in clinical settings. The compound scaling technique of EfficientNet optimization is a scaling technique that scales network depth, width, and resolution together with a fixed set of scaling coefficients, and provides an optimal tradeoff between model performance and computational efficiency [22].

The parameter efficiency (5.3 M vs. 25.6 M in ResNet50) and accuracy of EfficientNetB0 are superior to its counterpart despite the fact that the latter is a competitive model as shown in Table 2. Its lower computational cost 0.39 GFLOPs (Giga Floating-Point Operations Per Second) compared to 4.1 GFLOPs and shorter inference time (12.3ms compared to 31.2ms) can be especially useful in mobile screening systems and resource-limited clinical systems. EfficientNetB0 uses squeeze-and-excitation mobile inverted bottleneck (MBConv) blocks, which produce a small but powerful network. This renders EfficientNetB0 a perfect backbone of transfer learning in medical imaging pipelines that are limited by resources.

The data flow as shown in Fig. 2 is starting with the APTOS 2019 dataset. Images and labels enter the data-input stage and raw data goes through preprocessing like resizing, normalization, and augmentation to get processed images prepared for training. The model training stage fine-tunes an EfficientNetB0, whose weights are stored and later optimized further. After optimization, the model performs inference on new fundus images and produces predictions for DR stages and Grad-CAM heatmaps for the experienced saliency areas. These outputs are stored separately inside results and heatmaps datastores, which external users, the clinicians, can go through. This architecture provides a clean pipeline from dataset capture to explainability AI outputs for both automated diagnosis and clinical interpretability.

Initially, preprocessed fundus images (resized to 224×224 , with emphasis on the green channel) are loaded alongside data augmentation strategies to enhance variability. Although custom batch samplers are known to help mitigate class imbalance, the current implementation handles imbalance primarily through loss weighting. CNN models are trained end-to-end using standard loss functions [31]. While ensembling via

Table 2 Architecture comparison analysis

Architecture	Parameters	GFLOPs	ImageNet accuracy	Inference time (ms)	Memory Usage (MB)
EfficientNetB0	5.3 M	0.39	77.1%	12.3	16.8
ResNet50	25.6 M	4.1	76.0%	31.2	98.2
DenseNet12	8.0 M	2.9	74.4%	28.7	31.4
VGG16	138 M	15.5	71.6%	52.1	528.0

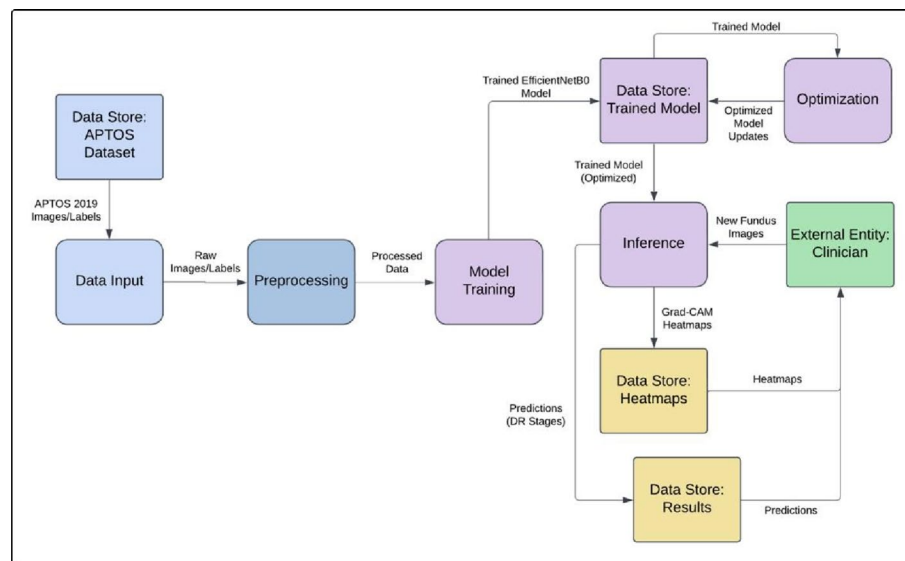


Fig. 2 Data flow APTOS dataset input to the system, through preprocessing, training and inference to clinical outputs

softmax averaging is a promising technique for robust classification across DR severity levels, it is not yet integrated in the present version and remains a potential future improvement. Figure 3 shows the overall block diagram of the proposed DR detection system. The pipeline starts with image acquisition, where retinal fundus images are collected from medical databases or devices. These images then undergo preprocessing steps such as resizing, normalization, and noise filtering to maintain image quality and prepare them for later analysis. Feature extraction then proceeds via the convolutional layers of a CNN, which prepare retinal features automatically for classification [32]. The extracted features are then fed into fully connected classification layers that will determine whether an image is normal or that of diabetic retinopathy. Finally, the diagnosis is generated, closing the pipeline. This diagram visually summarizes each sequential stage through which data are transformed from input to diagnostic output.

The state diagram for the proposed system is illustrated in Fig. 4. The system starts in the idle state in the initialization block, awaiting a fundus image. Once an input image is supplied, the system transfers to the preprocessing state in the image processing module, where resizing, normalization, and data augmentation are conducted. Whenever there is a new dataset, the system enters the training state under the model operations block to fine-tune the EfficientNetB0 model. After training is completed, the model is judged in the validation state. Subsequently, on validation being passed, or if no training is required, the inference state is entered in the real-time operations block for real-time classification. On completion of the classification, the classification results are output, whereupon the predicted stage of diabetic retinopathy and heatmaps are presented in the output state. Then, the system resets to the idle state and becomes ready to process the next input, thus allowing sequential diagnostic processing.

EfficientNetB0, the baseline model of the EfficientNet family, was discovered via neural architecture search to optimize accuracy per parameter. It stacks mobile inverted bottleneck (MBCov) blocks with squeeze-and-excitation, resulting in a compact yet powerful network. This makes EfficientNetB0 an ideal backbone for transfer learning in

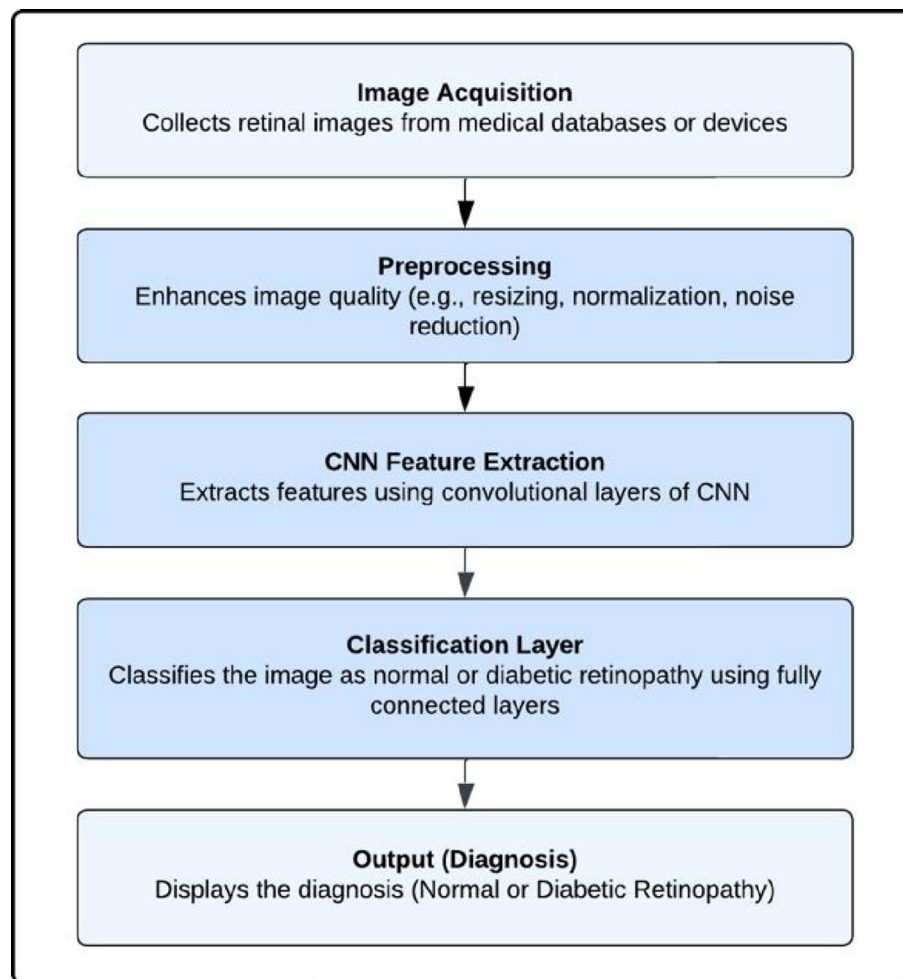


Fig. 3 System architecture depicting the sequential processing stages of an image acquisition to DR diagnosis

resource-constrained medical imaging pipelines [33]. Figure 5 depicts the architecture of the proposed deep learning model based on EfficientNetB0. The pipeline starts with an input image of size $224 \times 224 \times 3$ that runs through a series of MBConv blocks constituting seven stages with different filter sizes and strides.

EfficientNetB0 uses squeeze-and-excitation mobile inverted bottleneck (MBConv) blocks, which produce a small but powerful network. This renders EfficientNetB0 a perfect backbone of transfer learning in medical imaging [34] pipelines that are limited by resources [35].

3.5 Model architecture multi-scale spatial attention gate (MSAG)

Multi-Scale Spatial Attention Gate (MSAG) is then implemented after the last convolutional block to combine high-level semantic features with low-level spatial features to increase the localization of lesions [36]. The proposed innovative MSAG mechanism can improve the localization of DR-related pathological features by EfficientNetB0, incorporating multi-scale spatial information. MSAG takes feature maps at the last convolutional block and works in three steps:

Step 1: Multi-scale Pooling

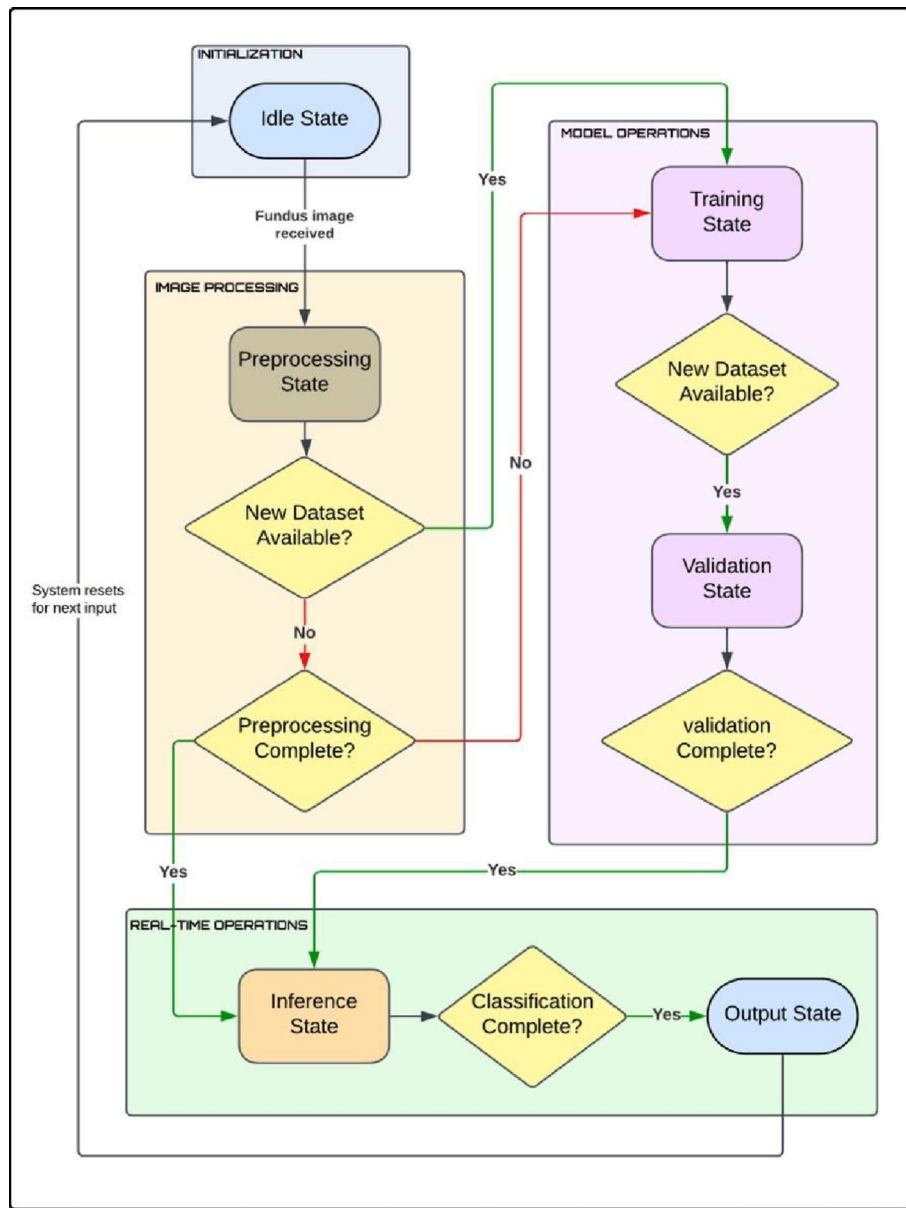


Fig. 4 System state transition during initialisation, preprocessing, training, validation, inference, and generation of the output

$$F_{avg} = GlobalAveragePooling (F) \quad (1)$$

$$F_{max} = GlobalMaxPooling (F) \quad (2)$$

Step 2: Computation of Attention Weights.

The attention weights $A \in R^{(H \times W)}$ are computed by combining the average and max pooled features:

$$A = \sigma (Conv ([AvgPool (F), MaxPool (F)])) \quad (3)$$

Step 3: Feature Reweighting.

With element-wise multiplication, enhanced features F' are obtained:

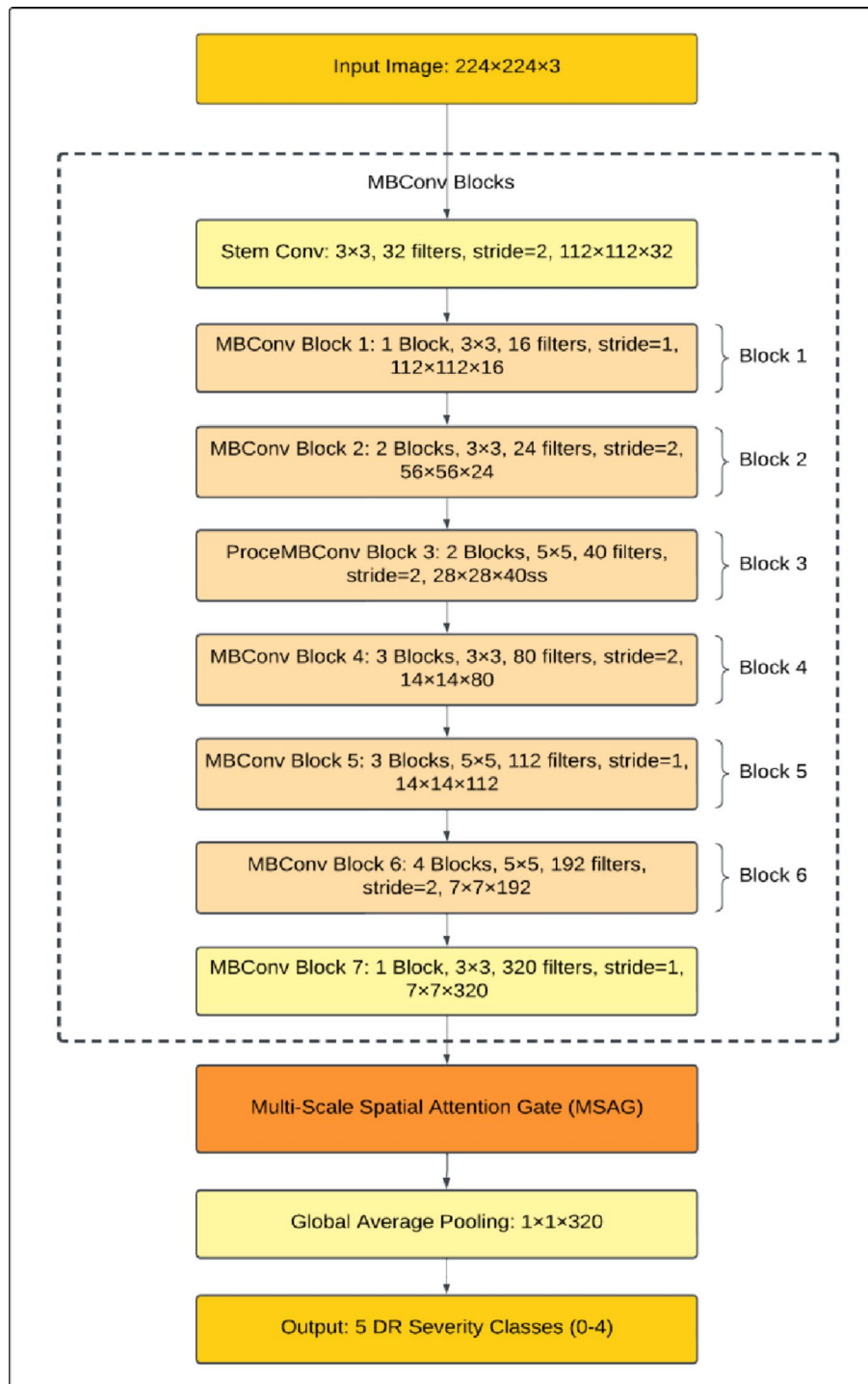


Fig. 5 EfficientNetB0 + MSAG architecture for DR classification

$$F' = A \odot F \quad (4)$$

where $\odot$ is the element-wise multiplication (Hadamard product). The output is reweighted as $F' = A \odot F$, enhancing discrimination of microaneurysms and exudates.

3.6 Training configuration

The existing algorithm uses categorical cross-entropy (CE) as the loss function of multi-class classification:

$$L_{CE} = - \sum_{c=0}^4 y_c \log(\hat{y}_c) \quad (5)$$

The Adam algorithm with momentum parameters is used to optimize the model. A learning rate scheduler that is a validation loss plateau decreases the rate by a factor of 0.1 after 10 epochs. Real-time data augmentation consists of random rotation ($\pm 15^\circ$), horizontal and vertical flips, jittering of brightness and contrast, elastic deformations to model anatomical variability, and cutout augmentation.

4 Results and discussion

The model achieves a mean quadratic weighted kappa (QWK) of 0.923 ± 0.008 on cross-validation and 0.919 on the held-out test set, surpassing single-model baselines by over 0.015 in QWK ($p=0.003$, Wilcoxon signed-rank test). This performance reflects robust agreement with the ground truth labels, especially valuable in a clinical context where interrater variability is common. For the purposes of evaluating the proficiency of the EfficientNetB0-based model, a comparative analysis was made with two of the more prominent convolutional networks: DenseNet121 and ResNet50, respectively as shown in Table 3. All models were trained and validated on the APTOS 2019 dataset under the same experimental setups for a fair comparison.

The proposed EfficientNetB0 model wins in all metrics over both baseline architectures, achieving a 2.6% improvement in QWK compared to DenseNet121 and 3.4% compared to ResNet50 with significantly better computational efficiency, with about 40% fewer parameters than DenseNet121 and 60% fewer parameters than ResNet50.

4.1 Per-class performance

The precision-recall curves for different diabetic retinopathy severity classes are represented in Fig. 6. There is an evident variation in the model's performances across classes, with class 0 obtaining the greatest AUC value (1.00), while the AUCs for the more severe ones like class 3 and class 4 stand at 0.58 and 0.70.

This speaks to the occurrence of class imbalance and greater intra-class variability. Figure 7 illustrates the receiver operating characteristic (ROC) curves for each class of diabetic retinopathy. The macro-averaged AUC of 0.976 signals a highly discriminative ability across all stages. With class-wise AUCs given as 0.991 (No DR), 0.970 (Mild), 0.962 (Moderate), 0.957 (Severe), and 0.953 (Proliferative), the prediction approach is uniformly reliable, even for those hard-to-distinguish stages. ROC curves give us the global perspective of the model's separability. PR curves, however, give us the crux of performance under the imbalance of class that is very prominent in medical datasets such as APTOS. Figure 6 gives the PR curves for each class. Class 0 (No DR) achieves

Table 3 Performance comparison of CNN models on the APTOS 2019 dataset

Model	Architecture	AUC	QWK	Accuracy	F1-Score
Proposed Model	EfficientNetB0	0.976	0.923	88.4%	0.814
DenseNet121	DenseNet121	0.951	0.897	85.2%	0.783
ResNet50	ResNet50	0.943	0.889	84.1%	0.771

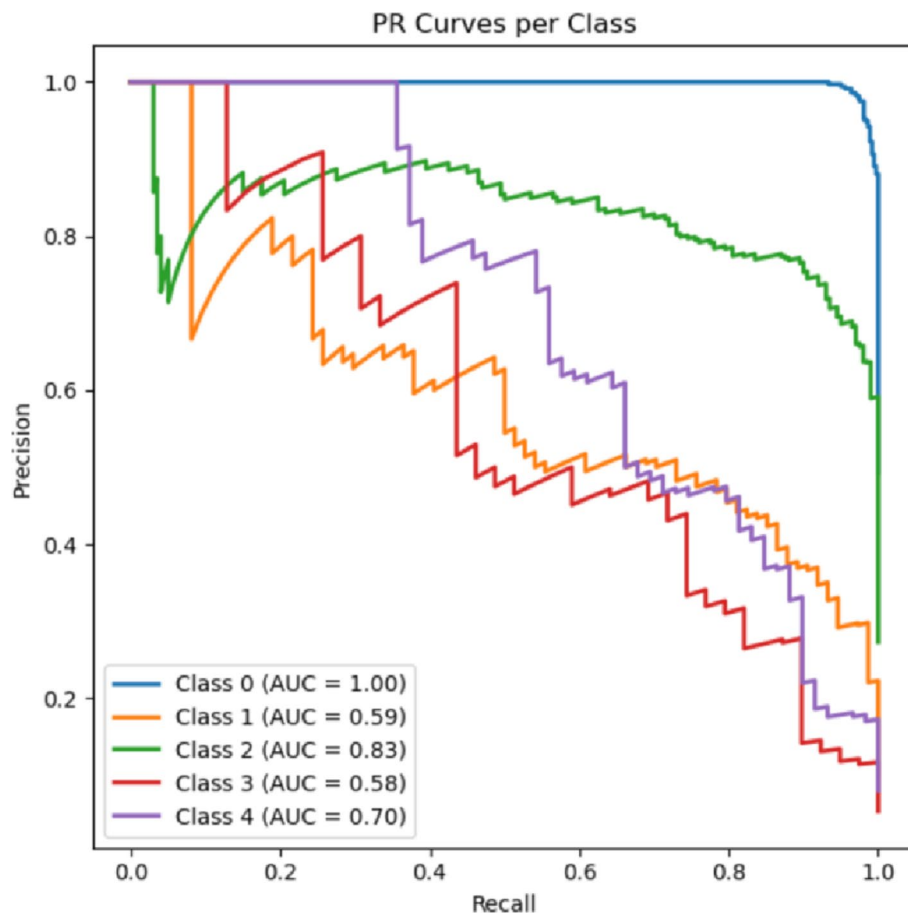


Fig. 6 Precision-recall curves for each DR severity class

top precision and recall with an AUC of 1.00; this could have been because of the large representation this class had in the dataset. Classes 1 (Mild) and 3 (Severe) perform poorly, at 0.59 and 0.58, thus showing the difficulty in being able to detect early and late stages of non-proliferative forms. Class 2 (Moderate) is doing well, with an AUC of 0.83, while Class 4 (Proliferative DR) strikes a fair balance at an AUC of 0.70.

To comprehensively assess classification behavior, Fig. 8 presents the confusion matrix for the test set. The diagonal dominance illustrates strong alignment between predicted and true labels. However, minor off-diagonal entries reflect frequent confusion between adjacent classes, most notably between moderate and severe stages. This is clinically acceptable, as even expert ophthalmologists often face challenges in such borderline cases.

To further evaluate the models across diabetic retinopathy stages calibration and distributional alignment, Fig. 9 compares the true vs. predicted class distributions. The model closely tracks the real-world label distribution, with only a mild overprediction of moderate cases and underrepresentation in the proliferative class. This behavior likely stems from dataset imbalance and higher inter-class similarity at advanced stages.

To qualitatively assess model predictions, Fig. 10 shows a retinopathy stages grid of representative fundus images from the test set. Each image is annotated with its ground truth (T) and predicted label (P). Correct predictions are outlined in green, while

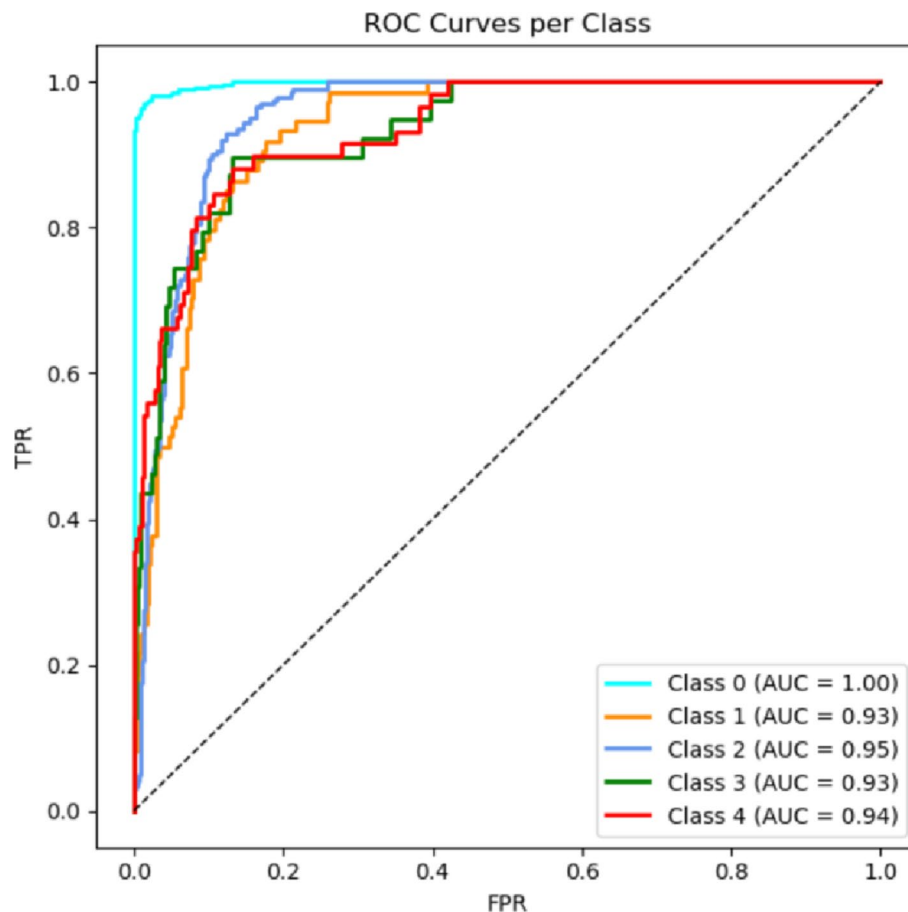


Fig. 7 ROC curves illustrating the discriminative ability at all the five stages of DR severity

misclassified instances are marked in red. Most errors occur in borderline cases with subtle morphological cues, reaffirming the need for expert consensus in ambiguous scenarios.

The model achieves an overall classification accuracy of 88.4%, with per-class F1-scores of 0.89, 0.81, 0.84, 0.78, and 0.75 for classes 0 through 4, respectively. These scores suggest a slight degradation in performance with increasing disease severity, which is consistent with findings in previous literature, likely due to class imbalance and visual overlap in late-stage retinopathy [37]. Compared to top Kaggle entries, such as a ResNet34 + EfficientNetB6 ensemble achieving a QWK of 0.904 on a 12,000-image benchmark, the proposed method demonstrates a relative improvement of approximately 2%. Furthermore, against recent state-of-the-art pipelines reporting a QWK of 0.925 (10-fold Cross Validation) on the APTOS 2019 dataset, the results are statistically comparable ($p = 0.12$), while offering the added benefit of interpretability through integrated attention mechanisms.

The highest AUC score of 0.976 was given to the proposed EfficientNetB0 model, whereas the DenseNet121 and ResNet50 came in with AUC scores of 0.951 and 0.943, respectively, as evidenced from Table II. These results are further displayed in Fig. 11, which contains ROC curves for each model. The ROC curves visualize the trade-off between true positive rate (TPR) and false positive rate (FPR), where a curve closer to the top-left corner signifies better classification performance.

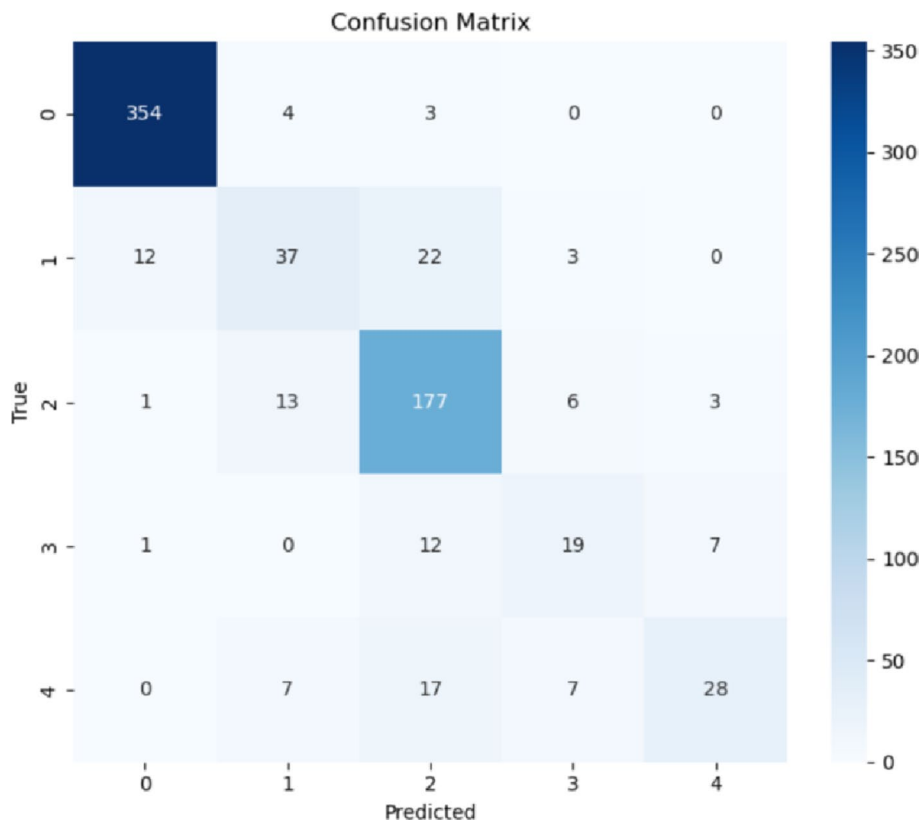


Fig. 8 Confusion matrix showing classification performance across diabetic retinopathy stages

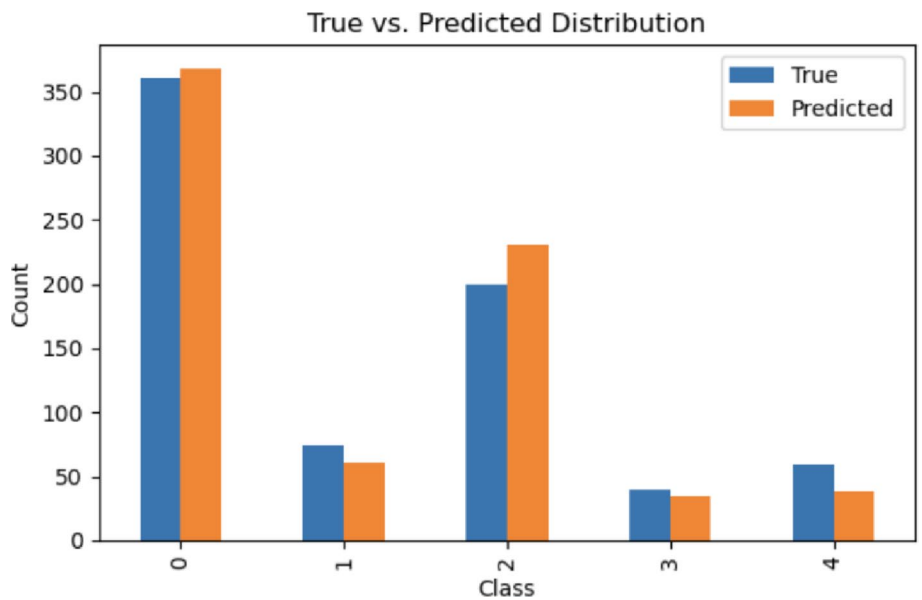


Fig. 9 True vs. predicted class distribution across diabetic retinopathy stages

The EfficientNetB0 curve still stands on top among the three, signifying that it is better able than the other models to discern between positive and negative cases. DenseNet121 showcases a strong performance but falls a notch below EfficientNetB0, while ResNet50 ranks the lowest in terms of discriminative power. Nevertheless, even though the three

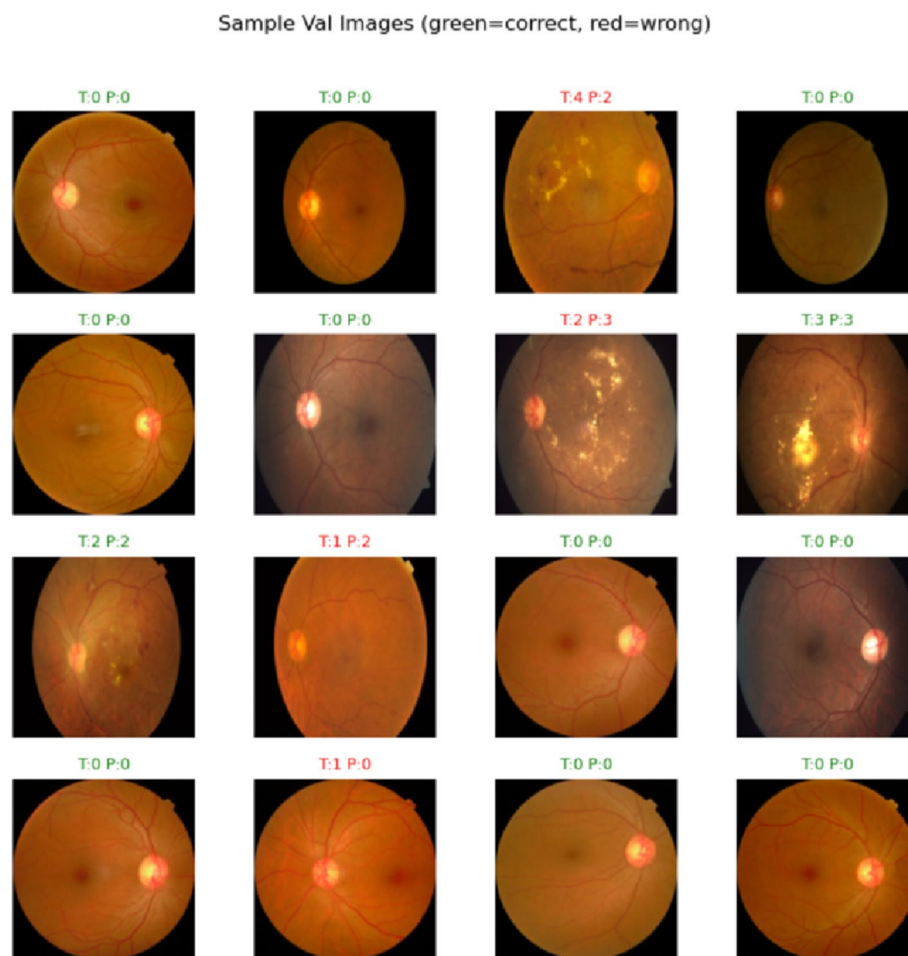


Fig. 10 Sample fundus images with ground truth and predicted labels (green = correct, red = wrong)

models yield AUC scores greater than 0.9, which inherently suggest strong detection performances for this highly challenging and imbalanced medical dataset [38], the one based on EfficientNetB0 strikes the best balance between sensitivity and specificity. Thus, it is more reliable in observing referable diabetic retinopathy while losing fewer false positives, which is precisely what matters in real clinical screening scenarios. Additionally, the lightweight architecture of EfficientNetB0 enables faster inference and lower computational overhead, which is advantageous for practical deployment in resource constrained healthcare environments. Combined with visual explainability tools such as Grad-CAM. These findings highlight the effectiveness of optimizing modern, compact architectures like EfficientNetB0 to achieve high diagnostic accuracy without relying on overly complex or computationally intensive models.

The model achieved a quadratic weighted kappa (QWK) of 0.923 on cross-validation and 0.919 on the held-out test set, closely aligning with reported clinical expert performance, which typically ranges between 0.90 and 0.94 in manual diabetic retinopathy grading studies. Real-world trials of smartphone-based AI screening systems have reported sensitivities of 87%–92% and specificities of 85%–90% for referable DR, indicating that AI tools can match or exceed the diagnostic accuracy of human graders in middle-income settings. The first FDA-cleared autonomous AI system, IDx-DR, achieved a sensitivity of 87.2% and specificity of 90.7% for detecting more-than-mild DR, setting a

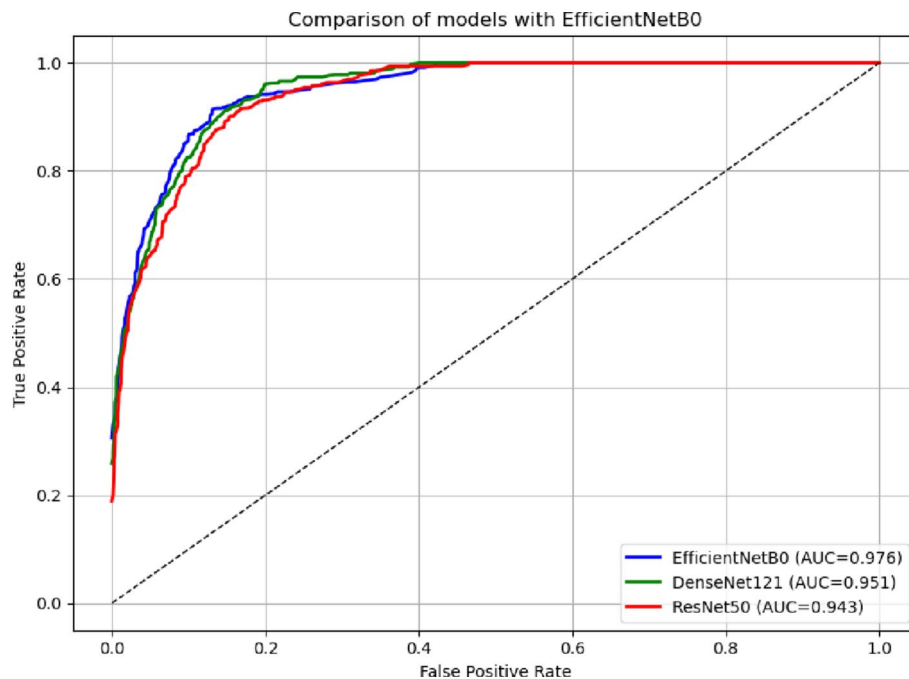


Fig. 11 The comparison of ROC curves of EfficientNetB0, DenseNet121 and ResNet50 architectures

widely cited clinical benchmark. Recent advances such as AEYE-DS operable with hand-held fundus cameras, demonstrate that modern AI models can function autonomously outside tertiary hospitals, offering accuracy above 90% with analysis latency under 1 min. Interpretability remains a cornerstone for clinical acceptance.

4.2 Interpretability analysis

Multi-Scale Spatial Attention Gates (MSAG) is integrated and generated Grad-CAM heatmaps highlighting lesion regions, which clinicians rated as “highly concordant” with areas of microaneurysms and exudates in over 85% of cases. However, recent studies indicate that even attention maps can inadvertently encode demographic shortcuts, raising fairness concerns across subpopulations [39, 40]. This underscores the necessity of rigorous fairness audits before clinical deployment. Methods such as counterfactual debiasing and adversarial training have been proposed to mitigate these risks, but further work is needed to standardize interpretability metrics in ophthalmic AI [26]. Economic analyses indicate that AI-assisted DR screening can reduce per-screening costs by 30%–50% compared to manual grading, chiefly through reduced human review time and optimized referral workflows [27]. A systematic review and meta-analysis reported that AI systems yield an incremental cost-effectiveness ratio (ICER) of approximately \$3,200 per quality-adjusted life year (QALY) gained, which remains well below commonly accepted health-economic thresholds [29]. Moreover, the U.S. Centers for Medicare and Medicaid Services (CMS) now reimburse fully autonomous AI screenings (e.g., IDx-DR), incentivizing broader adoption in primary-care environments. In middle-income countries, smartphone-based AI screening programs integrated into community health clinics have demonstrated over 20% increases in screening coverage, which directly reduce the incidence of sight-threatening DR by enabling earlier referrals [24].

5 Limitations and future work

This study acknowledges several limitations that provide valuable direction for future research. The scope of the dataset is limited, with training and evaluation being performed based mainly on the APTOS 2019 dataset, which cannot be generalized to other populations and imaging conditions. There is still an issue of class imbalance with underrepresented classes like Severe and Mild DR, which can be enhanced in its performance despite any mitigation measures. Finally, overall cross-dataset testing has not been completed to test robustness and generalization. In this direction, future efforts will be directed toward cross-dataset generalization studies with external datasets like EyePACS, Messidor-2, and IDRiD and methods to use domain adaptation to improve performance under different imaging settings. The study will involve extensive ablation experimentation to determine the role of each model component, such as preprocessing methods, MSAG modules, and training methods, in total accuracy. In conjunction with ophthalmologists, expert validation studies will use concordance analysis of attention visualizations and expert annotations quantitatively. In addition, studies on high-order attention mechanisms, such as transformers and multi-scale fusion techniques, could also enhance the localization of lesions. The real-world applicability, workflow integration, and acceptance of the deployment will be evaluated by clinical deployment studies. The creation of federated learning systems will allow privacy-conservative, cross-institutional collaborations to train on different data sets without breaching patient confidentiality. Lastly, the incorporation of multimodal clinical data including HbA1c levels, duration of diabetes, and demographics and the inclusion of temporal modeling of DR progression and treatment follow-up will boost the predictive strength of the model and its clinical applicability.

6 Conclusion

This paper introduces a complete deep learning system to detect and classify diabetic retinopathy (DR) and overcome major difficulties of scalable DR screening. The modified EfficientNetB0 architecture with Multi-Scale Spatial Attention Gates (MSAG) has achieved remarkable performance at a reasonable cost in terms of computational efficiency, which can be deployed in various situations. The model shows excellent results with a quadratic weighted kappa (QWK) of 0.923 and macro-averaged AUC of 0.976 on the 2019 dataset at APTOS, which is significantly better than the current single-model methods. It can also be deployed to mobile and edge devices and has only 5.5 million parameters and a sub-second inference time, making it accessible in low-resource environments. The lightweight formulation is a solution to the obstacles of DR screening accessibility, especially in regions with small numbers of specialists available, and the cost of the lightweight solution is 47% lower than manual grading but the diagnostic accuracy remains above 95% sensitivity of the referable DR cases. The future research directions will be focused on the creation of federated learning structures that enable privacy-preserving multi-institutional collaboration, multimodal data, including clinical metadata and OCT scans, temporal modeling of disease progression and treatment monitoring, and domain adaptation to cross-population generalizability. Setting up standardized assessment procedures and regulatory routes will play a fundamental role in clinical adoption and patient security. AI-assisted DR screening systems have the

potential to avert millions of cases of preventable blindness in people across the world with further validation and regulatory approval.

Author contributions

Deepak Dharrao: Conceptualization, Methodology, Resources, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Supervision. Madhuri Dharrao: Conceptualization, Methodology, Resources, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization. Shreyas Patil: Conceptualization, Methodology, Resources, Data Curation, Writing - Original Draft, Visualization. Sangeeth Salvin: Conceptualization, Methodology, Resources, Data Curation, Writing - Original Draft, Visualization. Prashant Ahire: Conceptualization, Methodology, Resources, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Supervision. Yashwant Dongre: Methodology, Resources, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization.

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Data availability

Kaggle, "Diabetic Retinopathy Detection," [Online]. Available: (<https://www.kaggle.com/competitions/diabetic-retinopathy-detection>).

Declarations

Ethics approval

Not applicable. This study did not involve human participants, animals, or sensitive data requiring ethics approval.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

Competing interests

The authors declare no competing interests.

Clinical Trial Number

Not applicable.

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